

4: Linear Regression:

$$\hat{y} = 6x + 10$$

ST₁ *
ST₂
ST₃
ST₄

Studied hrs	quiz Score
X	y
1	19
3	25
5	41
7	51

(A) $\hat{y} = 6 \times 9 + 10 =$
 $54 + 10 = \underline{64}$

(B) ~~ST₁~~ $y_{ST_3} = 41$

$\hat{y}_{ST_3} = 6 \times 5 + 10 = 40$

Residual of ST₃: $e = 41 - 40 = \underline{1}$

(C) ST₁: $y = 19$ $\hat{y} = 6 \times 1 + 10 = 16$ $e_1 = \underline{3}$

ST₂: $y = 25$ $\hat{y} = 6 \times 3 + 10 = 28$ $e_2 = \underline{-3}$

ST₃: $y = 41$ $\hat{y} = 6 \times 5 + 10 = 40$ $e_3 = \underline{1}$

ST₄: $y = 51$ $\hat{y} = 6 \times 7 + 10 = 52$ $e_4 = \underline{-1}$

~~ST₁~~ $3^2 + (-3)^2 + 1^2 + (-1)^2 = 11$ SSE

MSE = $\frac{11}{4} = \underline{2.75}$

RMSE = $\sqrt{2.75} = \underline{1.658}$

(D) SST:

$\bar{y} = (19 + 25 + 41 + 51) / 4 = 34$

$SST = (19 - 34)^2 + (25 - 34)^2 + (41 - 34)^2 + (51 - 34)^2$
 $225 + 81 + 49 + 289 = 644$

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